

DeFi SuperApp

Full-Stack Decentralized Protocol
Blockchain Technologies 2

- Option A capstone: AMM + lending + ERC-4626 yield vault.
- Chainlink-style oracle, OpenZeppelin Governor + Timelock, The Graph subgraph.
- Production posture: tests, security report, CI, deploy scripts, post-deploy checks.
- Instructor-approved 4-person team with separate ownership tracks.

Submission posture

Hardhat implementation with documented justification. Local demo and test suite are ready; final L2 verification requires live credentials.

Four Clear Ownership Tracks

Samat

AMM, factory, CREATE/CREATE2, LP accounting, swap math, gas inputs.

Dauren

lending, vault, oracle adapter, mocks, collateral and liquidation mechanics.

Arthur

governance, Timelock, UUPS treasury, vulnerability case studies, tests.

Ernar

frontend, subgraph, CI, deployment scripts, demo docs, slide delivery.

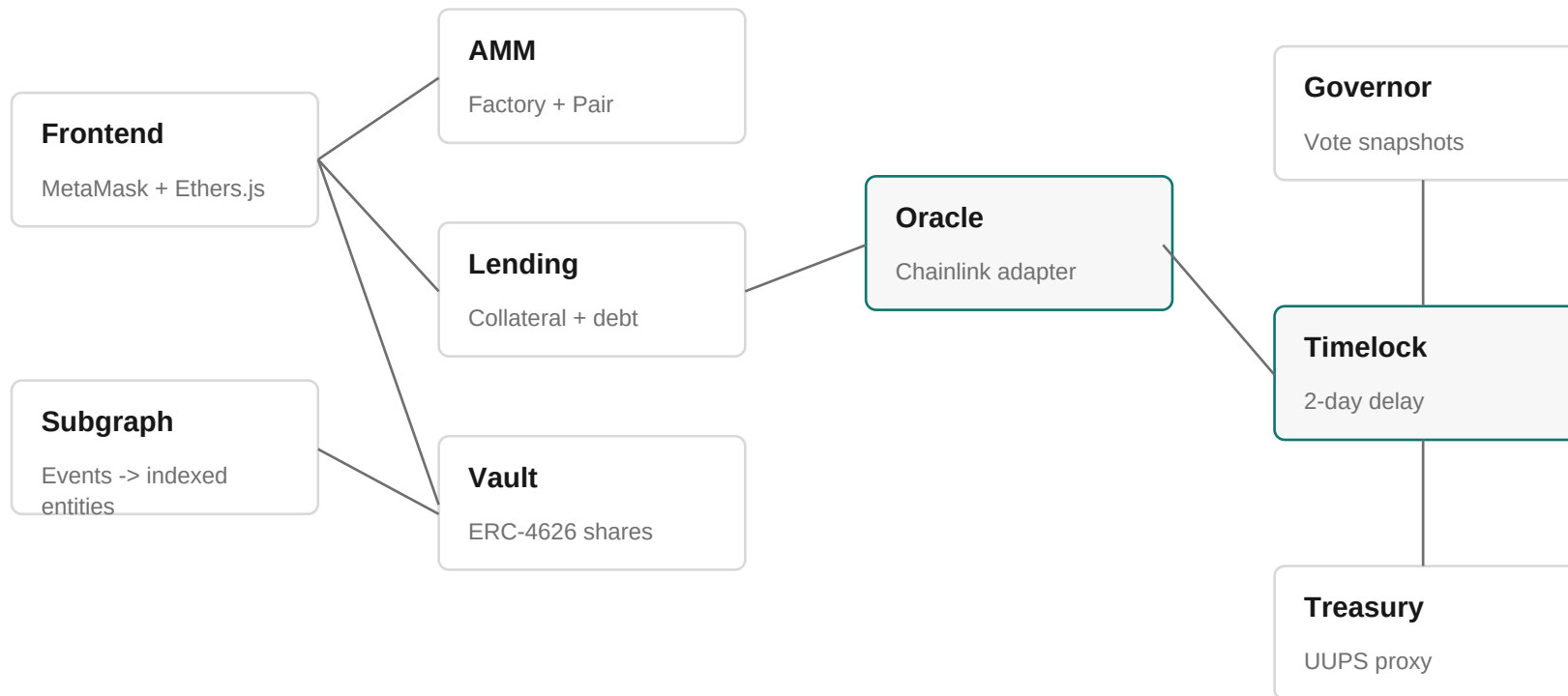
One Protocol, Three User Flows

- Swap assets through a constant-product AMM with 0.3% fee and slippage checks.
- Deposit stable assets into an ERC-4626 vault and receive tokenized shares.
- Use collateralized lending with LTV, health factor, liquidation, and linear interest.

Evidence

See README, docs/COVERAGE.md, docs/SECURITY_AUDIT.md, docs/GAS_REPORT.md, and scripts/postDeployCheck.js for reproducible checks.

Contract System Map



Mandatory Advanced Components

- UUPS upgrade path: ProtocolTreasuryV1 -> ProtocolTreasuryV2.
- Factory pattern: AMMFactory deploys pairs through CREATE and CREATE2.
- Assembly benchmark: AssemblyMath.sumYul compared to sumSolidity.
- Token standards: ERC20Votes + Permit, ERC-721 positions, ERC-4626 vault.

Evidence

See README, docs/COVERAGE.md, docs/SECURITY_AUDIT.md, docs/GAS_REPORT.md, and scripts/postDeployCheck.js for reproducible checks.

Controls And Case Studies

- Privileged functions use `AccessControl`; post-deploy check validates `Timelock` ownership.
- `ReentrancyGuard` and `CEI` are used where external calls or value transfers occur.
- Oracle adapter rejects stale, zero, negative, and incomplete round data.
- Before/after case studies cover reentrancy and missing access control.

Evidence

See `README`, `docs/COVERAGE.md`, `docs/SECURITY_AUDIT.md`, `docs/GAS_REPORT.md`, and `scripts/postDeployCheck.js` for reproducible checks.

87 Passing Tests

Tests

87 passing

Coverage

94.51% lines

Slither

0 High / Medium

- Core suite covers unit and revert paths for AMM, vault, lending, oracle, governance, treasury.
- 10 deterministic fuzz-style property tests.
- 5 invariant-style tests for k, ERC-4626 supply, lending accounting, treasury conservation, roles.
- 3 fork-style checks against mainnet USDC, Uniswap V2, and Chainlink ETH/USD.

94.51% Line Coverage

Tests

87 passing

Coverage

94.51% lines

Slither

0 High / Medium

- Hardhat coverage: 93.65% statements, 94.51% lines, 93.00% functions.
- Slither production-scope run: 0 High/Medium findings.
- CI compiles, tests, coverage-checks, lints, and runs Slither.

Minimal dApp Demo

- MetaMask connection and chain detection.
- Reads balance, voting power, delegate, AMM reserves, vault shares, and loan position.
- Write actions: swap, deposit, vote.
- Human-readable errors for missing wallet, wrong network, rejected transactions, and failed calls.

Evidence

See README, docs/COVERAGE.md, docs/SECURITY_AUDIT.md, docs/GAS_REPORT.md, and scripts/postDeployCheck.js for reproducible checks.

Reproducible L2 Path

- Deploy script supports Arbitrum Sepolia, Optimism Sepolia, and Base Sepolia.
- Post-deploy script checks Timelock delay, Governor settings, treasury roles, and no admin backdoor.
- Final live explorer verification requires funded key, RPC URL, and explorer API key.
- Subgraph builds locally and is ready for Graph Studio deployment after live addresses are known.

Evidence

See README, docs/COVERAGE.md, docs/SECURITY_AUDIT.md, docs/GAS_REPORT.md, and scripts/postDeployCheck.js for reproducible checks.

What To Show In 15 Minutes

- Run local Hardhat node and deploy mocks.
- Open the minimal frontend at <http://127.0.0.1:5173/>.
- Connect wallet, inspect account state, perform swap/deposit/vote flows.
- Run npm test, npm run coverage, npm run slither, and post-deploy checks.

Evidence

See README, docs/COVERAGE.md, docs/SECURITY_AUDIT.md, docs/GAS_REPORT.md, and scripts/postDeployCheck.js for reproducible checks.